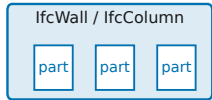


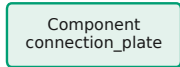
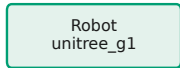
## (a) IFC design model



IFC GUID  
2A9nYhQvT5Q9B7m7X1pA01

*design intent only*

## (b) assets pass alone



*geometry, collision, mass OK*

## (c) composed scene fails



missing\_workzone  
payload



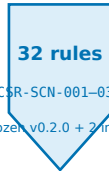
wrong\_up\_axis



unsupported  
generated\_friction

*cross-asset faults (fault taxonomy)*

## (d) scene-level contract



30 frozen v0.2.0 + 2 in v0.3.0

- ✓ closed references
- ✓ provenance
- ✓ task preconditions